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Sharing responsibility for the good☆

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ABSTRACT

Relative to individual decision-making, decision-makers in markets have been argued to be more willing to accept negative externalities of their actions. A key mechanism potentially explaining this phenomenon is a diffusion of responsibility between actors in markets. In the preregistered experiment reported here, we test for an effect of responsibility diffusion in a new context: economic transactions with positive externalities. In particular, we test if participants' willingness to pay for the vaccination of a child against measles differs between individual and bilateral decision-making. We find no such effect. Our study thus adds an(other) instructive null-result to the literature trying to pin down precisely which aspects of market interactions affect precisely which types of morally relevant decisions.

1. Introduction

When you are deciding whether nor not to buy that fashionable sweatshirt, that tasty chocolate, that fancy new smartphone—any product for which you can be pretty sure that it was produced under dubious circumstances: what shapes your decision? An obvious factor is price: relative to an 'ethically sound' substitute product, the dubious one might need to be cheaper for you to agree to the deal. Several social considerations could shape your decision too. Once you know that others are buying the product as well, you might feel less pivotal in causing the product's negative externalities. Relatedly, you might use a replacement argument, i.e., you might say: "if I don't buy that product now, somebody else will do so later, so the damage is done anyway".

Considerations like these are central in the ongoing discussion about whether the allocation of goods and services through market mechanisms makes the involved individuals' decisions less likely to be in line with these individuals' own and/or their societies' moral standards or not (Bowles, 1998; Bruni & Sugden, 2013; Dufwenberg, Heidhues, Kirchsteiger, Riedel, & Sobel, 2011; Sandel, 2012; Satz, 2010; Shleifer, 2004; Storr & Choi, 2019). Over the last decade, experimental economists have contributed to this debate by breaking down selected aspects of economic transactions into decision situations that can be studied under controlled laboratory conditions. A series of papers, e.g., has looked into the effects of different forms of markets on 'moral erosion' (Falk & Szech, 2013; Kirchler, Huber, Stefan, & Sutter, 2016; Sutter, Huber, Kirchler, Stefan, & Walzl, 2020; Ziegler, Romagnoli, & Offerman, 2021). Others have looked at 'ethical consumption' by consumers and its effects on producers more directly (Pigors & Rockenbach, 2016a, 2016b; Rode, Hogarth, & Le Menestrel, 2008).

Arguably the most widely known experimental work on the topic is Falk & Szech's 2013 study "Markets and Morals" (Falk & Szech, 2013)—which has probably also received most criticism (Breyer & Weimann, 2015; Lütge & Rusch, 2013) and provoked most follow-up work (Bartling, Fehr, & Özdemir, 2021; Bartling & Özdemir, 2021; Falk, Neuber, & Szech, 2020; Irlenbusch & Saxler, 2019). In that paper, Falk & Szech find that participants' individual willingness to accept ('WTA') in stylized trades that cause morally problematic externalities is higher in individual decision contexts compared to social ones (bilateral and multilateral trading in their case)—i.e., individual participants agree to trades that cause negative externalities only at higher prices relative to participants in social decision contexts. The two main effects they report are of medium size (treatment comparisons reported in their Figure 1, page 707: 0.8 > Cohen's h's > 0.5). Falk & Szech offer two main explanations for their findings: (*i*) social contexts provide information about others' WTA and participants could use this information to adjust their own standards downwards, thus serving their own monetary interests at a cost to their own moral standards—a conjecture compatible with recent findings by Ziegler et al. (2021); (*ii*) the responsibility for causing the negative externality is diffused between participants in the social contexts, thus alleviating the 'moral burden' felt by the individual. It is this second conjecture that we take up in this paper.

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Our study complements earlier works that also looked into effects of responsibility diffusion on 'moral erosion' in trading contexts (Bartling et al., 2021; Falk et al., 2020; Falk & Szech, 2013; Irlenbusch & Saxler, 2019; Kirchler et al., 2016). In short, that literature arrives at mixed results. For the sake of parsimony and given the currently still small number of relevant previous studies, we refrain from interpreting and trying to consolidate their results here and refer the reader to Bartling et al. (2021), Falk et al. (2020), Irlenbusch and Saxler (2019) for thorough literature overviews. Instead, we add new evidence from a preregistered experiment that provides a critical test of whether the potential effect of responsibility diffusion observed in some studies generalizes to a new context: transactions with positive externalities. More precisely, we test for a possible main effect of changing the number of responsible persons involved in decisions that have positive externalities for third parties while keeping the material consequences for everyone involved equal. To do so, we build on the designs of previous studies (Falk & Szech, 2013; Irlenbusch & Saxler, 2019), but change the externality's valence from negative to positive.

2. Experimental design

2.1. Dependent variable and treatments

The central dependent variable of our experiment is participants' willingness to pay (WTP) for the vaccination of one child against measles, administered via UNICEF—as also used by Kirchler et al. (2016), Sutter et al. (2020), Ziegler et al. (2021). Thus, relative to previous work on the topic, we 'reverse' the decision context. Earlier work investigated economic transactions that, relative to some fixed status quo, benefit the involved agent(s) monetarily (modeling realized gains from trade) but cause negative externalities. Here, instead, we look at economic transactions that are costly for the involved agent(s), relative to a fixed status quo, but cause a positive externality. Examples for such transactions include the purchase of 'ethical' or 'green' products, which are more expensive than regular products, but for which customers can be pretty sure that part of the revenue is used to finance social or environmental goals, i.e., which create positive side-effects for third parties.

In essence, thus, previous work focused on how much participants' would need to be offered (in terms of monetary gains) in order for them to accept a negative externality, i.e., their critical willingness to accept (WTA) for causing 'bad things'. Our focus, accordingly, is on participants' WTP for 'good things'. Note that reversing only one of the two characteristic features of the decision context—type of externality and monetary consequence for the actor—would trivialize the problem: Who would not be willing to accept higher gains when that causes a positive externality in addition? And who would be more prone to engage in a transaction that, relative to the status quo, yields a loss *and* causes a negative externality? Thus, the non-trivial 'reversed' question is: how much are participants willing to pay for creating a positive externality? Therefore, this WTP is what we use as the dependent variable in our experiment.

In order to test for an effect of diffused responsibility then, we experimentally manipulate the number of decision-makers. In this regard, we draw on the setups used by Irlenbusch and Saxler (2019) (their treatments 'NEUTRAL NO INFO' vs. 'INDIVIDUAL') and Falk and Szech (2013) (their 'individual condition' vs. the 'first control condition' they discuss on page 709). Our baseline is individual decision making (treatment 'IND'). Here, participants are assigned an endowment, E, and need to decide how much they would be willing to pay, P_i, in order for the positive externality to be created, resulting in a payoff of $\pi_i(P_i) = E - P_i$. For clean comparability with our bilateral decision-making treatment ('BIL'), i.e., to rule out any effects induced by social preferences, we match a passive second participant, j, to every active participant, i, in IND who receives the same payoff like i, thus $\pi_j(P_i) = E - P_i$ in treatment IND. Passive participants make no payoff relevant decisions and this is common knowledge—we elicit hypothetical WTPs and expectations about the active participants' WTPs from the passive participants, though. We elicit P_i using the well-known Becker–DeGroot–Marschak (BDM) mechanism (Becker, Degroot, & Marschak, 1964).

The bilateral decision-making treatment is identical except that the second participant j now also makes a decision: we elicit both participants' WTPs, P_i and P_j, and then use the smaller one, $\min(P_i, P_j)$, as the WTP in a joint BDM mechanism for both participants. This diffuses responsibility between the two as follows. In IND, P_i definitively fixes the upper limit on the cost of creating the externality. In BIL, however, i's WTP is not decisive in case $P_j < P_i$ and vice versa. Thus, when making their decisions, participants in BIL do not know whether they will be pivotal or not. Furthermore, analogously to a higher WTA precluding more negative externalities in previous work, a lower WTP precludes more positive ones in our setup.

It is worth noting that we follow Falk and Szech (2013) and Irlenbusch and Saxler (2019) in conceptually considering both participants involved in the transaction as 'first parties' regarding the externality. Maximizing responsibility of one participant in IND requires making the other one fully passive, which results in a degenerated type of 'transaction' between the two in that treatment. However, in line with previous studies, we do not consider the effect of the active participant's decision on the passive one in IND an (additional) externality.[1]

2.2. Experimental procedures

Data were collected in March 2018. We recruited participants through Amazon's 'Mechanical Turk' platform (mturk.com). They completed an online survey collecting all independent as well as the dependent measures. The full survey used as well as a full description of the rationale of this study were preregistered prior to data collection via the AEA RCT Registry (DOI: 10.1257/rct.2751-2.0). Our research question was: will the effects reported by Falk & Szech and others for negative externalities generalize to positive externalities? We had no hypothesis on whether we would find a positive, negative, or null effect. The full experimental instructions are also available as an online supplement to this paper.

Step 1: Diligence check. In order to exclude participants who were not following our experimental instructions carefully, the survey began with a diligence check. Participants had to copy, letter by letter, a meaningless string of characters backwards and in all uppercase letters. Based on the results of previous studies, we expected about 25% of participants to fail this test (in fact we had 23% such cases). These participants were blocked from further participation in the study.

Step 2: Demographics. Participants then answered a brief demographics questionnaire asking for gender (male/female/other), age (in years), subjective socio-economic status (on a ladder with nine rungs), and number of siblings.

Step 3: Background information on measles. In order to make sure that all participants are informed about the prevalence and health risks of measles, participants were asked to carefully read a summary of the official fact sheet on measles provided by the WHO. Participants then needed to answer three comprehension questions concerning the fact sheet summary. After this, participants were informed about how measles vaccinations were to be provided by the researchers (through a donation to UNICEF after data collection), about the current price of two doses of measles vaccine (required for full protection of one child; $0.74 at the time of data collection), and receipts for the researchers' previous donations to UNICEF were shown in order to maximize credibility of the instructions.

[1] We thank an anonymous reviewer for bringing this point to our attention.

Step 4: General instructions. Participants were informed about the matching mechanism and payment procedures used in the study (post-hoc matching with delayed payment after all decisions by all participants had been collected, donations to UNICEF were also made after the completion of data collection). Afterwards, they were required to correctly answer a comprehension question concerning these procedures.

Step 5: Randomization, treatment instructions, and decisions. Participants were randomized into either IND, BIL, or assigned the passive role in IND. In IND, the active participant made a decision (also see below). This participant's decision was implemented and affected both themselves as well as their matched passive participant. In BIL, two participants actively made decisions and the joint result of their decisions was implemented, affecting both of them. All decisions made, i.e., in both IND and BIL, also had a consequence for a third party (each decision either resulted in a donation of the amount needed for a complete measles vaccination of a child in need to UNICEF by the researchers or not, also see below).

Participants read the instructions for the respective treatment (and role) that they were randomly allocated into. Afterwards they had to correctly answer several comprehension questions in order to be able to proceed. Decisions in the treatments were as follows.

In IND, half of the participants were randomly allocated to the passive role. These participants were informed about the decision that the active participants would be making. They were then asked to make a hypothetical decision in the role of an active participant, but were informed upfront that this decision would have no monetary (or other) consequences.

The active participants made the following decision. They received an endowment of $2.50. Then, they were asked to indicate the maximum amount that they are willing to pay (WTP) out of this endowment in order for a child in need to receive full vaccination against measles. We used the BDM method (Becker et al., 1964) to elicit this WTP, i.e., if a randomly drawn 'price', between $0.00 and $2.50, for the vaccination was smaller than a participant's WTP, the amount needed for the vaccination was donated to UNICEF by the researchers and the participant received their endowment minus that price. Else, the participant received their full endowment, but no vaccination was donated. Payoff consequences for passive participants were exactly the same as for their respective matched active decision maker.

In BIL, the two randomly matched active decision makers made exactly the same decision as the active participants in IND. However, the donation to UNICEF was made only if both participants' WTPs were greater than or equal to the randomly drawn price for the vaccination.

Step 6: Expectations. After the decision stage, we elicited all participants' beliefs about the WTP of their matched partners. I.e., we asked participants to estimate how much their respectively matched other participant was (or 'would have been' for matched passive partners) willing to pay for the measles vaccination. Belief elicitation was not incentivized.

Step 7: Final questionnaire. Participants answered two trolley scenarios (Foot, 1967; Greene, 2008) and completed the moral foundations questionnaire (Graham et al., 2011).

Step 8: Debriefing. Participants were thanked and informed about when they would receive their show-up fee and bonus payment (i.e., the payoff resulting from their decisions), and where to find proof of the researchers' donations to UNICEF. Eventually, participants' decisions resulted in 75 complete measles vaccinations for children through UNICEF and we posted the respective donation certificate on one of our personal homepages.

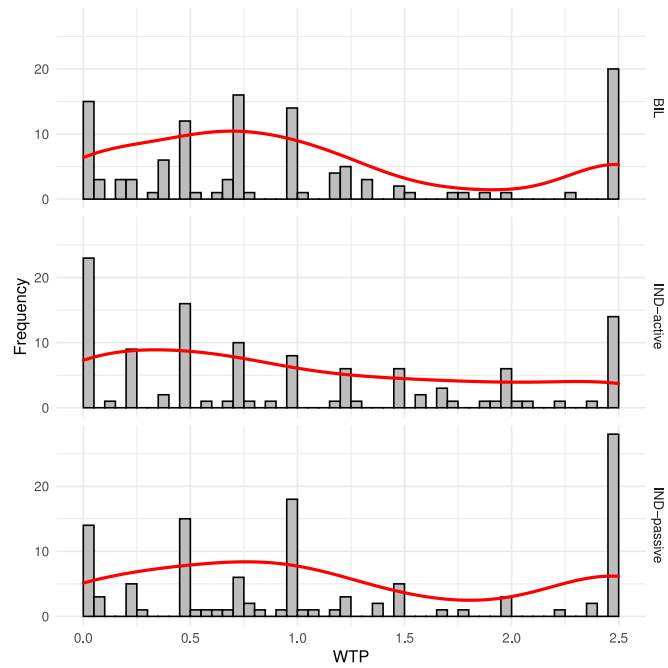

Fig. 1. Willingness to pay (WTP) by treatment (role); in red: smoothed and scaled density curves; note that WTPs in IND-passive are hypothetical.

Preregistered exclusions and robustness checks. Apart from the exclusion of participants who failed the diligence check, we preregistered to run all primary analyses on (*a*) all data and (*b*) the subset of participants who stated a WTP larger than $0.00 and smaller than $2.50. Results from a pilot with $N = 30$ participants indicated that irrespective of the interval used to elicit participants' WTP, a fraction of all participants always stated $0.00, i.e., these participants are not willing to pay any positive amount for the vaccination of a child through UNICEF, and another group of participants always stated that they are willing to pay any positive amount within the range used to elicit the WTP (we tested upper boundaries of $2.50, $5.00, $7.50, and $10.00). Based on the data from our pilot, we expected roughly 20% of participants to choose one of these two extreme ends of the WTP-range. In fact, that share turned out to be 28% in the full data collection.

3. Results

We aimed to recruit 120 participants per treatment (and role), yielding a targeted sample size of 360 (120 participants in IND-active, 120 in IND-passive, and 120 in BIL). This sample size is a compromise between experimental costs and statistical precision which we adjusted so as to be able to detect effects similar in size to those reported by Falk & Szech (see 'Introduction'). Given this sample size and for significance level $\alpha = 0.05$, our main treatment comparisons are sufficiently powered ($1 - \beta = 0.8$) for effect sizes down to $d = 0.38$ when using a two-tailed t-test or a two-tailed Wilcoxon–Mann–Whitney-test. Eventually, i.e., after exclusions due to failure to pass the diligence test, we have 358 participants in our sample. Table 1 shows descriptive statistics.

Main treatment comparisons. Fig. 1 shows the distribution of WTP by treatment (role). A two-tailed Wilcoxon–Mann–Whitney-test indicates no significant difference in WTP between IND-active and BIL ($W = 6,928, p = 0.69$; two-tailed t-test: $t = -0.11, p = 0.92$, estimated effect size $d = 0.01$). The difference between the hypothetical WTP elicited in IND-passive and the pooled actual WTP in IND-active and BIL reaches borderline statistical significance ($W = 16,030, p = 0.049; t = 1.95, p = 0.053$, estimated effect size $d = 0.23$). Our null-result for differences between IND-active and BIL is robust when controlling for the available

Table 1
Descriptive statistics, overall and broken down by treatment(-role). Note that WTPs in IND-passive are hypothetical.

Variable	*Total*	*IND-active*	*IND-passive*	*BIL*
N	358	119	119	120
Gender (M/F/O)	148/209/1	51/68/0	46/72/1	51/69/0
Mean age	39.32	39.29	39.05	39.62
Subjective SES	5.37	5.31	5.47	5.33
Mean WTP (SD)	1.06 (0.85)	0.99 (0.84)	1.19 (0.89)	1.00 (0.82)

demographic variables. Table 2 shows the results of OLS-regressions using available controls. Furthermore, an equivalence test using the TOST procedure (Lakens, 2017) indicates that the presence of a true effect larger than $d = 0.4$ is very unlikely given our data ($\alpha \leq 0.05, 1 - \beta \geq 0.8$, 90%-CI for the mean difference of WTP between BIL and IND-active: [-0.166, 0.189]). I.e., our results indicate that if responsibility diffusion does have an effect in our context, that effect is very likely notably smaller that the effects originally reported by Falk and Szech (2013).

A possible concern about these results is that our participant's WTP choices follow unusual distributions with choices clustering noticeably at the upper and lower bounds of the WTP range. In fact, we have 30, 34, and 36 such 'extreme choices' (WTP $\in \{\$0.0, \$2.5\}$) in BIL, IND-active, and IND-passive, respectively—a phenomenon also observed in other laboratory studies on decision making with morally laden outcomes (Ockenfels, Werner, & Edenhofer, 2020; Sutter et al., 2020).

Based on our pilots, we expected such distributions and therefore also preregistered a comparison of choices that are 'interior' ($\$0 < \text{WTP} < \$2.5$). Comparing these using two-tailed Wilcoxon–Mann–Whitney- and t-tests indicates a borderline significant difference in WTP between IND-active and BIL ($W = 4,438, p = 0.066$; two-tailed t-test: $t = 2.32, p = 0.022$, effect size $d = 0.36$, achieved power $1 - \beta = 0.65$.) Note that this results indicates higher 'interior' WTP in IND-active ($M = \$0.98, SD = \$0.62$) relative to BIL ($M = \$0.78, SD = \$0.48$). Table 3 shows the pertaining OLS-regression results.

When comparing the distributions of extreme choices in IND-active and BIL, we also obtain a borderline significant result (Fisher's exact test, $OR = 0.36, p = 0.049$, achieved power $1 - \beta = 0.52$). This result, however, indicates higher 'extreme' WTP in BIL ($20 \times \$2.5$ and $10 \times \$0.0$) relative to IND-active ($14 \times \$2.5$ and $20 \times \$0.0$). Due to the low statistical power of these analyses, we refrain from interpreting these results further here. Nonetheless, given their repeated occurrence across studies (Ockenfels et al., 2020; Sutter et al., 2020), we think that future work should try to look specifically into drivers of such extreme decision behavior.

Beliefs. Fig. 2 plots the distributions of all beliefs about the matched participant's WTP we elicited and the respectively stated own WTP. Pairwise Wilcoxon–Mann–Whitney- and t-tests indicate no significant differences between any of these distributions (all p-values > 0.10), i.e., our participants' beliefs were not systematically biased up- or downwards (Ziegler et al., 2021).

Incidental findings. As it turns out, we observe a significant main effect for gender (see Fig. 3; pooling across treatments BIL and IND-active: $N_{Men} = 102, N_{Women} = 137, M_{Men} = \$0.85, SD_{Men} = \$0.80, M_{Women} = \$1.11, SD_{Women} = \$0.83, t = -2.44, p = 0.015$, estimated effect $d = 0.32$, achieved power $1 - \beta = 0.78$; $W = 5,574, p = 0.007$). OLS estimates from Table 2 indicate that women's WTP is about \$0.18 higher than men's in the (non-restricted) sample. Model 3 of Table 2 also shows that there is no significant interaction effect of treatment and gender. We explored several (non-preregistered) other ways of analyzing the data, all leading to the same qualitative results with the gender effect being the only robust finding.

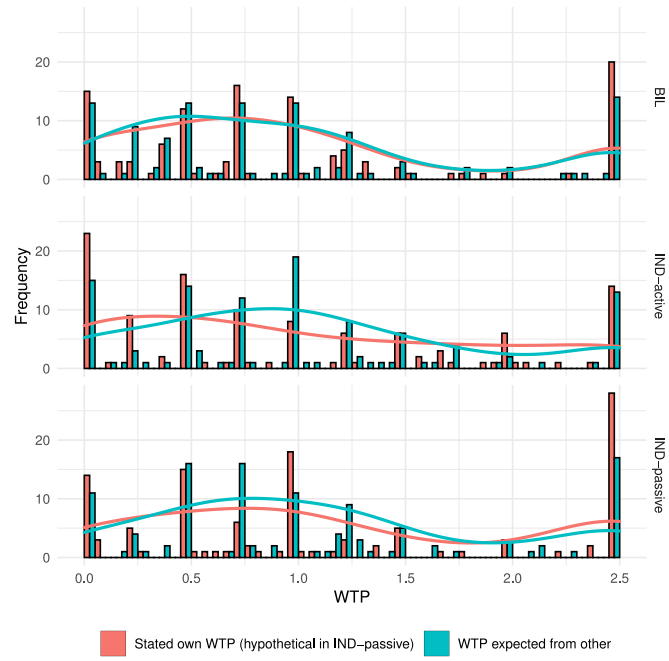

Fig. 2. Beliefs about other participants' WTP by treatment (blue) and stated own WTP (red) by treatment(-role); stated own WTP is payoff relevant in IND-active and BIL and hypothetical in IND-passive. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

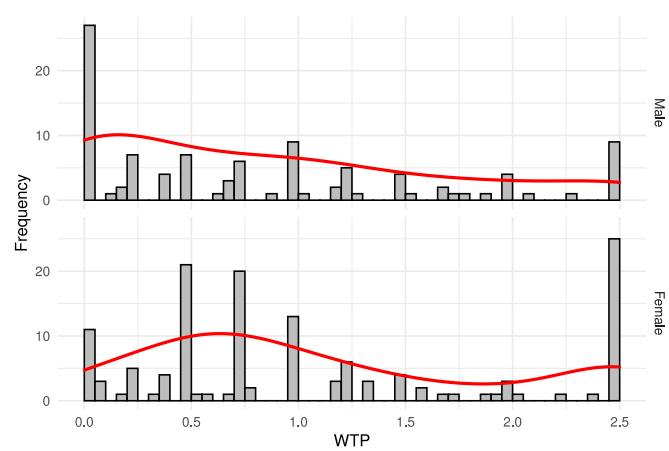

Fig. 3. Willingness to pay (WTP) in IND-active and BIL by gender (one observation identifying as "Other" is dropped); in red: smoothed and scaled density curves.

4. Discussion and conclusion

In the preregistered experiment reported here, we tested for an effect of diffused responsibility in the context of economic transactions that cause positive externalities. Particularly, we tested if participants would be willing to pay more (or less) for the vaccination of

Table 2
OLS regressions exploiting all available demographic controls. The dependent variable is WTP. (One observation identifying their gender as "Other" is dropped.)

	Model 1	Model 2	Model 3
BIL	0.01	0.01	0.21
	(0.11)	(0.11)	(0.36)
IND-passive	0.19*	0.18*	0.63*
	(0.11)	(0.11)	(0.37)
Gender		0.18**	0.32**
		(0.09)	(0.16)
Age		0.01	0.01
		(0.004)	(0.004)
Subj. SES		−0.02	−0.02
		(0.03)	(0.03)
BIL × Gender			−0.13
			(0.22)
PAS × Gender			−0.28
			(0.22)
Constant	0.99***	0.59**	0.38
	(0.08)	(0.26)	(0.32)
Observations	357	357	357
R^2	0.01	0.03	0.03
Adjusted R^2	0.004	0.02	0.01
Residual std. error	0.85 (df = 354)	0.84 (df = 351)	0.85 (df = 349)
F statistic	1.77 (df = 2; 354)	2.16* (df = 5; 351)	1.77* (df = 7; 349)

Note:
*$p < 0.1$.
**$p < 0.05$.
***$p < 0.01$.

Table 3
OLS regressions on sample restricted to non-extreme choices (WTP $\neq \{0.0, 2.5\}$) exploiting all available demographic controls. (One observation identifying their gender as "Other" is dropped.)

	Model 1	Model 2	Model 3
BIL	−0.19**	−0.20**	−0.12
	(0.08)	(0.09)	(0.29)
IND-passive	−0.09	−0.09	0.15
	(0.09)	(0.09)	(0.29)
Gender		−0.04	0.02
		(0.07)	(0.12)
Age		−0.002	−0.002
		(0.003)	(0.003)
Subj. SES		−0.01	−0.01
		(0.02)	(0.02)
BIL × Gender			−0.04
			(0.17)
PAS × Gender			−0.15
			(0.18)
Constant	0.98***	1.17***	1.07***
	(0.06)	(0.20)	(0.25)
Observations	258	258	258
R^2	0.02	0.02	0.03
Adjusted R^2	0.01	0.01	0.0002
Residual std. error	0.56 (df = 255)	0.56 (df = 252)	0.56 (df = 250)
F statistic	2.65* (df = 2; 255)	1.26 (df = 5; 252)	1.01 (df = 7; 250)

Note:
*$p < 0.1$.
**$p < 0.05$.
***$p < 0.01$.

a child against measles when making this decision individually vs. in anonymous pairs.

It stands to reason that our manipulation of the decision context could affect a difference in WTP: participants could, e.g., expect their matched partner to have a higher WTP than themselves and therefore adjust their WTP upwards so as not to prevent their partner from being able to cause the externality also at higher prices. Analogously, expecting the partner to have a lower WTP could lead to a downward adjustment. However, we do not observe any robust effect in one or the other direction.

Thus, our results add another caveat to the existing literature on responsibility diffusion in morally laden economic decision contexts—which has produced mixed results so far (Bartling et al., 2021; Falk et al., 2020; Falk & Szech, 2013; Irlenbusch & Saxler, 2019; Kirchler et al., 2016). Our (null-)results indicate that any effects possibly induced by diffused responsibility in the context of transactions with negative externalities do not generalize straightforwardly to transactions with positive externalities. Instead, our findings indicate that decision making in this context is quite heterogeneous (with about 28% of participants stating 'extreme' WTPs) and shaped by individual characteristics (with gender showing a medium effect in our sample).

Naturally, our study has several limitations. We did our best to exclude non-diligent participants, but still our study is subject to the remaining constraints of online experiments. We also did our best to maximize the credibility of our donation procedures, but still some participants may have been sceptical given that other studies on MTurk often use deception. Here, the usual argument applies: While both factors may add some noise to our data, randomization of participants into treatments should still result in meaningful treatment differences if the experimental manipulation has an effect. Moreover, we followed previous work in how we operationalized externalities on third parties and diffusion of responsibility, particularly Falk et al. (2020), Falk and Szech (2013), and Irlenbusch and Saxler (2019). Nonetheless, using vaccinations to model externalities may invoke attitudes and norms quite specific to the vaccination context and manipulating the pivotality within groups of decision-makers to model a diffusion of responsibility for those externalities may not fully capture what our participants feel to be responsible for. Thus, it is up for debate and further scrutiny whether these operationalizations capture all relevant aspects of the real-world decision contexts we are interested in.

In sum, nonetheless, we consider our experiment to be a good first attempt at testing for possible effects of diffused responsibility in the context of transactions with positive externalities. Our main statistical analyses are well-powered for effect sizes down to $d \approx 0.4$. Smaller effects might of course be there, but our experiment indicates that robust effects of the size of those reported in the influential study by Falk and Szech (2013) are not obtainable in this domain of morally laden economic decision making.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.socec.2022.101953.

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